Particle advection simulation

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Abstract

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1 Introduction

2 Description

Calculations consist of two parts:

1. preparation of the vector fields, which includes reading the data files and reconstruction of the smooth fields in space and time,

2. simulation of advection of particles driven by the reconstructed fields. Particles are propagated with explicit Euler method.

3 Interpolation of the Spatial Field

In this section, we describe the interpolation of velocity fields obtained from HYCOM (for ocean currents) and Copernicus (for winds). The data is stored as multidimensional tables in NCDF format. We will not delve into the tedious process of reading the data files. After preprocessing, we obtain rectangular grids of eastward and northward components of the velocity fields: $u_j(t)$ and $v_j(t)$. The coordinates are degrees of latitude φ and longitude λ. Here, j is the spatial index of the grid point. The actual sorting does not matter because we build a kd-tree for nearest neighbor search. It is crucial that the positions of grid points remain constant over time. We exclude grid

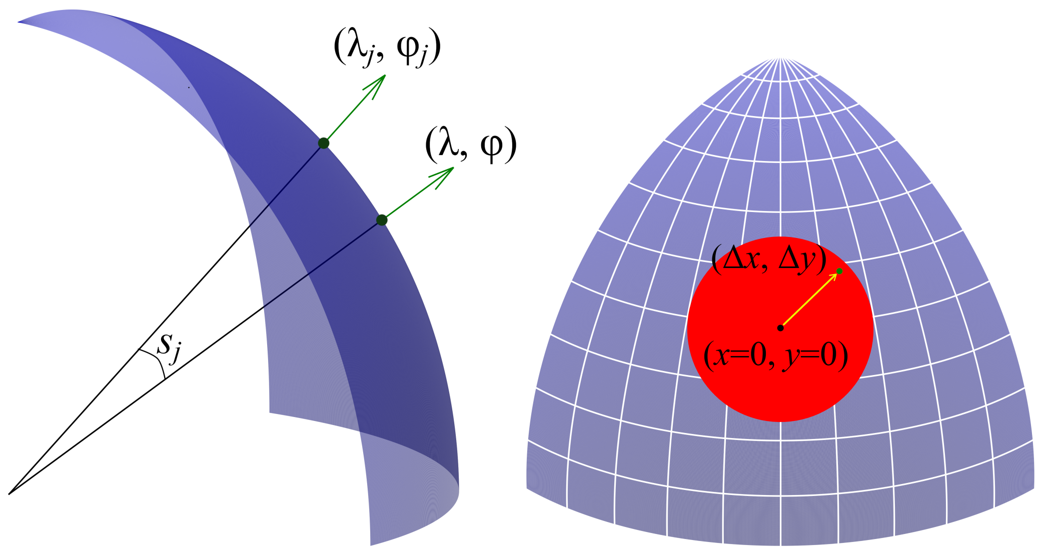

Figure 1: Left panel: Great Circle Distance between the query point $(\lambda,\ \varphi)$ and the grid point $(\lambda_j,\ \varphi_j)$. Right panel: Equidistant azimuthal projection onto a plane tangent at the query point

points on land to improve simulation speed and save memory. The variable t represents time; it is discrete in the data but interpolated with cubic splines to achieve continuous dependence on time.

Simulating on large scales necessitates accounting for the Earth's curvature, because any flat projection introduces distortions in distances, angles, or areas, or all of these aspects together. We utilize the fact that the Earth's surface is a smooth manifold and use local azimuthal equidistant projections for short-distance advection to minimize distortions.

The position of a particle, referred to as the query point, is used as the origin for local projections. At any given time t, the query point (λ, φ) is projected to minimize distortions in distances, calculated as Great Circle Distances (in radians):

$$s_j = \tan^{-1} \frac{|\bar{n} \times \bar{n}_j|}{\langle \bar{n} \cdot \bar{n}_j \rangle} = \tan^{-1} \frac{\sqrt{1 - (\sin\varphi \sin\varphi_j + \cos\varphi \cos\varphi_j \cos(\lambda - \lambda_j))^2}}{\sin\varphi \sin\varphi_j + \cos\varphi \cos\varphi_j \cos(\lambda - \lambda_j)}, \tag{1}$$

where s_j is the Great Circle Distance between the query point (λ, φ) and a grid point j with lat-lon coordinates (λ_j, φ_j), $\bar{n}$ and $\bar{n}_j$ are normal unit

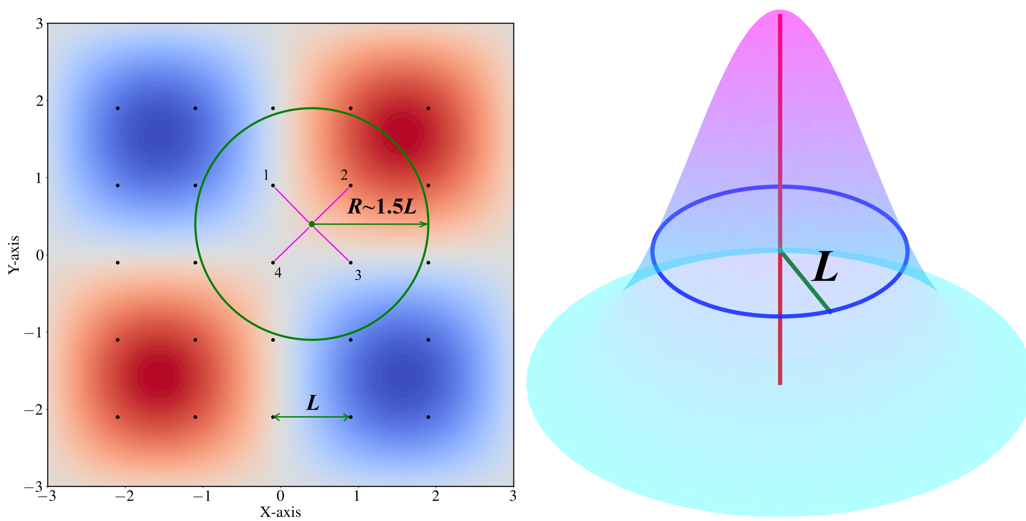

Figure 2: Left panel: Grid and a radius of search of nearest grid points. Right panel: Gaussian weight function

vectors at these points.

We interpolate field values between grid points by calculating a weighted average of the values at the nearest grid points. Weight functions decrease with the distance from the query point. Nearest neighbors are searched on a kd-tree built at the start of the simulation. The search radius is chosen such that $R \approx 1.5L$, where L is the grid step, ensuring that there are at least 4 grid points within the search radius for any typical query point position.

The weight function is defined as:

$$\mu(s_j) = \exp\left(-\frac{s_j^2}{L^2}\right), \tag{2}$$

where s_j is the distance from the query point to the grid point with index j. L^2 governs the shape of the Gaussian. Through trial and error, it is found that the shape must relate to the grid step L for better approximation. If the query point coincides with the grid point, the weight is 1. Weights for grid points further away decrease smoothly with distance. The normalized weight - or a measure of grid point j in the approximation is then

$$\rho_j = \frac{\mu(s_j)}{\sum_{s<R} \mu(s)}, \tag{3}$$

where the sum in the denominator is carried out for every grid point within the search radius.

The interpolated values of the eastward and northward components of the vector field are then

$$u(\lambda, \varphi, t) = \sum_{s_j < R} \rho_j u_j(t), \quad v(\lambda, \varphi, t) = \sum_{s_j < R} \rho_j v_j(t). \tag{4}$$

4 Numerical solution for tracers

Let's wrap it up: we have grids of velocity components $u_j(t)$ and $v_j(t)$ (in m/s), but the grid itself is on a sphere, the coordinates of grid points are in degrees of latitude and longitude. We have tracer equations on a flat surface, with coordinates in meters:

$$\begin{aligned} \dot{x} &= u(x, y, t), \\ \dot{y} &= v(x, y, t). \end{aligned} \tag{5}$$

Now we describe the simulation for one tracer particle. We get the spherical coordinates of the tracer – the query point: λ and φ, search for its nearest neighbors and interpolate the values $u(\lambda, \varphi, t)$ and $v(\lambda, \varphi, t)$. We construct the tangent plane at the query point, it becomes the origin of Euclidian frame: $(x = 0, y = 0)$.

We perform an explicit Euler step:

$$\begin{aligned} \Delta x &= \Delta t \cdot u(\lambda, \varphi, t), \\ \Delta y &= \Delta t \cdot v(\lambda, \varphi, t), \end{aligned} \tag{6}$$

New coordinates of the tracer become $(x' = \Delta x, y' = \Delta y)$ in meters on the tangent plane at $t' = t + \Delta t$. We project them back to the spherical coordinates:

$$\begin{aligned} \lambda' &= \lambda + \tan^{-1} \frac{\Delta x \cdot \sin \Delta s}{R\Delta s \cdot \cos\varphi \cos \Delta s - \Delta y \cdot \sin\varphi \sin \Delta s}, \\ \varphi' &= \sin^{-1} \left(\sin\varphi \cos \Delta s + \frac{\Delta y}{R\Delta s} \cos\varphi \sin \Delta s \right), \end{aligned} \tag{7}$$

where $R = 6378000m$ is the Earth radius,

$$\Delta s = \frac{\sqrt{(\Delta x)^2 + (\Delta y)^2}}{R} \tag{8}$$

is Great Circle Distance between the previous and the new position of the tracer.